

Multi-Commodity Energy Market Data

Notebook 1 of the Cross-Commodity Energy Trading analytics suite.

This notebook characterises the statistical behaviour of nine European energy commodities (crude oil, natural gas, carbon allowances, German and Nordic power, coal, refined products, and foreign exchange) from January 2020 through the present. The sample starts in 2020 because EEX publishes EUA auction reports from that year onward, which binds the common-date panel.

Executive summary

A European energy trading desk monitors crude oil (Brent), natural gas (TTF), carbon allowances (EUA), baseload power (Germany and Nord Pool), coal (API2), gasoline (RBOB), gasoil, and EUR/USD. These nine instruments span the fossil fuel, power, carbon, and currency markets that interact through the merit order, fuel-switching economics, and EU ETS compliance.

The data reveal three stylised facts that matter for risk management. First, TTF natural gas is the most volatile instrument in the complex, with a coefficient of variation exceeding 100% and rolling annualised volatility routinely above 30%. This is structural, not incidental: gas markets lack the global fungibility of crude oil, so a regional supply shock such as a pipeline outage or a storage constraint transmits directly into price rather than being absorbed by seaborne arbitrage. Second, every commodity except EUR/USD rejects normality in a Jarque-Bera test. The fat tails visible in the return histograms mean that a Gaussian VaR model calibrated to these data will systematically understate tail risk. Third, the price paths diverge over the sample: German power and carbon end well above where they started, while gas and coal spend much of the period below their starting levels. That divergence is consistent with a tightening carbon price, though a six-year window cannot separate policy from the 2022 supply shock that dominates the middle of the sample.

For a commercial trading desk these patterns are not academic. An integrated energy trading book spans crude, gas, and power, and its risk is driven by exactly these volatility and tail-risk characteristics. Understanding the distribution of each commodity is the necessary first step before constructing spreads, modelling correlations, or measuring portfolio risk, the subjects of Notebooks 2 through 4.

1. Market microstructure

The nine instruments in this dataset are not an arbitrary collection. Each has a specific market structure, trading venue, and role in the European energy complex.

Microstructure matters here because it sets how quickly information reaches the price: an exchange-traded futures contract and a weekly auction do not respond to the same news on the same day, which shows up later in the correlation estimates.

Trading venues

Commodity	Benchmark	Primary Exchange	Contract	Unit
Crude Oil	Brent	ICE Futures Europe	Futures + CFD	USD/bbl
Natural Gas	TTF	ICE Endex	Futures, spot	EUR/MWh
Carbon	EUA	ICE / EEX	Futures, auction	EUR/t CO2
German Power	DE Baseload	EEX	Futures (Phelix)	EUR/MWh
Nordic Power	Nord Pool System	Nasdaq Commodities	Futures	EUR/MWh
Coal	API2	ICE Futures Europe	Futures	USD/tonne
Gasoline	RBOB	CME (NYMEX)	Futures	USD/gallon
Distillate	ULSD Heating Oil	CME (NYMEX)	Futures	USD/gallon
FX	EUR/USD	CME / OTC	Futures, spot	—

How desks map onto these instruments

A typical integrated energy trading floor is organised into books that map onto these instruments directly. A **crude and refined products** desk trades Brent-linked grades and product futures, managing the crack spread, the margin between crude input and product output, against a refining position. A **gas and power** desk trades TTF, NBP, and European power, managing spark and dark spreads as gas-fired and coal-fired generation compete in the merit order. A **carbon** desk manages EUA positions for both compliance surrender and trading, with direct exposure to the EU ETS allowance price. The panel assembled here is deliberately chosen to span all three.

Regulatory data context

Under **REMIT II** (Regulation 2024/1106), wholesale energy market participants must report transactions to ACER, including OTC trades. The price data flowing through this notebook is the same class of data that feeds into ACER's market surveillance. The daily frequency used here matches the standard reporting granularity for organised market trades (T+1).

2. Data pipeline

The dataset is assembled from three sources into a DuckDB star schema.

- **yfinance**: Brent (BZ=F), RBOB (RB=F), Gasoil (custom), EUR/USD (EURUSD=X)
- **ENTSO-E Transparency Platform**: German day-ahead baseload and Nord Pool system prices via REST API
- **carbon-ets**: EUA auction clearing prices from EEX primary market reports (2020–2026)

All prices are normalised to EUR/MWh using standard conversion factors: Brent at 1.628 MWh/bbl and API2 coal at 6.978 MWh/tonne. The coal factor is worth stating explicitly: API2 is CIF ARA coal specified at 6,000 kcal/kg NAR, which gives 6.978 MWh/tonne. The 8.141 MWh/tonne figure that circulates widely belongs to a tonne of coal equivalent at 7,000 kcal/kg, a different contract specification, and using it would overstate coal's energy content by 17%, flattering the dark spread throughout.

USD-quoted series are converted through EUR/USD before the energy conversion. The EURUSD=X quote is USD per one EUR, so a USD price is *divided* by it. Refined products stay in USD/bbl for the crack spread rather than going to EUR/MWh: RBOB is converted from USD/gallon at 42 gallons per barrel, and ICE Gasoil from USD/tonne at 7.45 barrels per tonne. The DuckDB `fact_prices` table holds the normalised panel.

The pipeline runs idempotently: each fetch checks the latest date in the database and appends only new observations. Configuration is managed through Hydra (`config/pipeline.yaml`), so that unit conversions, tickers, and date ranges live in version-controlled configuration rather than in notebook cells.

```
In [1]: import sys
from pathlib import Path
sys.path.insert(0, str(Path.cwd().parent / "src"))

import duckdb
import pandas as pd
import numpy as np
import plotly.graph_objects as go
from plotly.subplots import make_subplots
from scipy import stats as sp_stats

# Report theme colours
NAVY = '#00003C'
OFFWHITE = '#FAFAFA'
TEAL = '#2E7D6F'
RED = '#C44536'
GRAY = '#6B6B6B'
COLORS = [TEAL, RED, '#6C8EBF', '#D4A843', '#8B6C9E', '#4A9C8C', '#C47E3B',

conn = duckdb.connect(str(Path.cwd().parent / 'energy_data.db'), read_only=True)

prices = conn.execute(''
    SELECT date, commodity_key, price_eur_mwh
```

```

        FROM fact_prices
        ORDER BY date, commodity_key
    ''').df()

wide = prices.pivot(index='date', columns='commodity_key', values='price_eur')
comm_names = {
    'BRENT': 'Brent Crude', 'TTF': 'TTF Gas', 'EUA': 'EUA Carbon',
    'DE_POWER': 'DE Baseload', 'N01_POWER': 'Nordic N01', 'API2': 'API2 Coal',
    'RBOB': 'RBOB Gasoline', 'GASOIL': 'ULSD Heating Oil', 'EURUSD': 'EUR/USD'
}

print(f'Loaded {len(prices):,} rows across {wide.shape[1]} commodities')
print(f'Date range: {wide.index[0].date()} to {wide.index[-1].date()}')
print(f'Trading days: {len(wide)}')
conn.close()

```

Loaded 16,120 rows across 9 commodities
 Date range: 2020-01-01 to 2026-08-07
 Trading days: 2411

3. Summary statistics

Per-commodity descriptive statistics: count, mean, standard deviation, minimum, maximum, skewness, excess kurtosis, and the coefficient of variation ($CV = \sigma/\mu$), which provides a scale-free volatility measure comparable across instruments with different price levels.

A formal Jarque-Bera test is reported for each commodity. The null hypothesis is that the returns are normally distributed. A p-value below 0.05 rejects normality, which happens for eight of the nine series here; EUR/USD is the only one that does not. The implication for risk measurement is that any model assuming Gaussian returns (e.g., a simple variance-covariance VaR) will misprice tail risk.

```

In [2]: def compute_stats(series):
        s = series.dropna()
        jb_stat, jb_p = sp_stats.jarque_bera(s)
        return pd.Series({
            'Count': len(s),
            'Mean': s.mean(),
            'Std': s.std(),
            'CV': s.std() / s.mean(),
            'Min': s.min(),
            'Max': s.max(),
            'Skewness': s.skew(),
            'Kurtosis': s.kurtosis(),
            'JB Stat': jb_stat,
            'JB p-val': jb_p,
        })

stats_df = wide.apply(compute_stats).T
stats_df.index = [comm_names.get(c, c) for c in stats_df.index]
display(stats_df.round(4))

```

```
# Highlight rejections of normality
print("\nJarque-Bera normality test (H0: normally distributed):")
for idx, row in stats_df.iterrows():
    verdict = "REJECT normality" if row['JB p-val'] < 0.05 else "FAIL TO REJ
    print(f" {idx:20s}: JB={row['JB Stat']:8.1f}, p={row['JB p-val']:.2e}
```

	Count	Mean	Std	CV	Min	Max	Skewness	Kurtosis	
API2 Coal	1504.0	17.5247	11.7232	0.6690	4.9929	57.8160	1.6888	2.1621	10
Brent Crude	1660.0	41.5982	11.5757	0.2783	10.9283	72.3512	-0.0888	-0.0864	
DE Baseload	2411.0	103.9038	86.2862	0.8304	-53.8708	699.4417	2.4977	8.6232	95
EUA Carbon	1438.0	64.8449	20.4169	0.3149	14.6000	97.5100	-0.9154	-0.1986	2
EUR/USD	1717.0	1.1167	0.0568	0.0509	0.9596	1.2341	-0.1368	-0.6704	
ULSD Heating Oil	1660.0	2.4951	0.8150	0.3267	0.6104	5.1354	0.2594	-0.1332	
Nordic NO1	2411.0	76.2905	74.0828	0.9711	-6.8138	660.0604	2.5088	9.9046	123
RBOB Gasoline	1660.0	2.2408	0.6540	0.2918	0.4118	4.2762	0.0834	0.4758	
TTF Gas	1659.0	49.9200	44.6523	0.8945	3.5100	339.1960	2.3266	6.5581	44

```
Jarque-Bera normality test (H0: normally distributed):
API2 Coal      : JB= 1003.4, p=1.30e-218 → REJECT normality
Brent Crude    : JB= 2.7, p=2.55e-01 → FAIL TO REJECT
DE Baseload    : JB= 9938.5, p=0.00e+00 → REJECT normality
EUA Carbon     : JB= 202.8, p=9.01e-45 → REJECT normality
EUR/USD        : JB= 37.6, p=6.68e-09 → REJECT normality
ULSD Heating Oil : JB= 19.9, p=4.86e-05 → REJECT normality
Nordic NO1     : JB= 12335.4, p=0.00e+00 → REJECT normality
RBOB Gasoline  : JB= 17.2, p=1.80e-04 → REJECT normality
TTF Gas        : JB= 4445.9, p=0.00e+00 → REJECT normality
```

TTF gas records the highest coefficient of variation, exceeding 100%, more than four times that of Brent crude. The Jarque-Bera statistic for TTF is in the thousands, with a p-value indistinguishable from zero. This reflects the propensity of gas markets to experience sharp dislocations: pipeline outages, storage constraints, and seasonal demand swings each produce moves of several standard deviations. Coal (API2) and RBOB gasoline show similar patterns, positive skewness and excess kurtosis, indicating markets where upside shocks dominate over the sample period.

EUR/USD is the sole instrument that plausibly passes the normality test, with near-zero excess kurtosis and a Jarque-Bera p-value above conventional thresholds. This is consistent with the behaviour of a deep, liquid currency pair. In this book it is a scaling factor for dollar-denominated contracts rather than a primary risk driver.

4. Normalised price paths

Each commodity rebased to 100 at the start date. Normalisation removes scale differences, crude at 70 USD/bbl and power at 50 EUR/MWh would otherwise be incomparable on a single axis, and lets relative performance stand out directly.

```
In [3]: normed = wide.bfill(axis=1) / wide.bfill(axis=1).iloc[0] * 100

fig = go.Figure()
for i, col in enumerate(normed.columns):
    fig.add_trace(go.Scatter(
        x=normed.index, y=normed[col],
        mode='lines', name=comm_names.get(col, col),
        line=dict(color=COLORS[i % len(COLORS)], width=1.2),
    ))

fig.update_layout(
    title=dict(text='Normalised Price Paths (Start Date = 100)', font=dict(c
    xaxis=dict(title='', gridcolor='#E0E0E0'),
    yaxis=dict(title='Index (100 = start)', gridcolor='#E0E0E0'),
    plot_bgcolor=OFFWHITE, paper_bgcolor=OFFWHITE,
    legend=dict(orientation='h', y=-0.25),
    height=550, margin=dict(l=50, r=50, t=50, b=100),
    hovermode='x unified',
)
fig.show()
```



Gasoil and Brent track each other closely through 2023, consistent with crude as the dominant feedstock cost, and the crack spread between them oscillates around a relatively stable mean. German power trends upward over the sample, alongside rising carbon costs and the phase-out of nuclear and coal capacity under the Energiewende. TTF starts near multi-year lows, spikes through 2021 into the 2022 supply crisis, then falls back from 2023 onward as European storage refills and LNG import capacity expands. Coal and gas spend most of the sample below their starting levels.

The gaps between these paths are the spreads Notebook 2 decomposes. The gap between power and gas is the spark spread; the Brent-gasoil gap is the crack spread. Carbon enters both, through the emission cost embedded in the generation margin.

5. Log returns distribution

Log returns $r_t = \ln(P_t/P_{t-1})$ per commodity, with a fitted normal distribution overlaid. The gap between the histogram and the normal curve reveals the presence

and magnitude of fat tails, the statistical signature of markets where extreme moves occur more often than a Gaussian model predicts.

For a risk manager, this visual gap is the difference between a VaR breach once every 100 days (as the normal distribution implies at the 99th percentile) and the more frequent breaches that actually occur. Notebook 4 handles this excess tail mass in the marginals, through GARCH filtering and empirical residual quantiles, rather than in the dependence structure: the fitted copula on this panel turns out to be close to Gaussian.

```
In [4]: from omegaconf import OmegaConf

from energy_cross_commodity.risk.returns import compute_log_returns
from energy_cross_commodity.utils.config import load_config

# Power prices clear negative in oversupply, where a plain log ratio is
# undefined. The displacement keeps those days in the sample instead of
# quietly dropping the most extreme observations in the series.
displacements = OmegaConf.to_container(
    load_config().risk.price_displacement_eur, resolve=True
)
log_returns = compute_log_returns(wide, displacements)

n_cols = 3
n_rows = 3
commodities = list(wide.columns)

fig = make_subplots(rows=n_rows, cols=n_cols,
    subplot_titles=[comm_names.get(c, c) for c in commodities],
    vertical_spacing=0.08, horizontal_spacing=0.05)

for idx, col in enumerate(commodities):
    row = idx // n_cols + 1
    c = idx % n_cols + 1
    r = log_returns[col].dropna()
    mu, sigma = r.mean(), r.std()
    x = np.linspace(r.min(), r.max(), 200)
    pdf = sp_stats.norm.pdf(x, mu, sigma)

    fig.add_trace(go.Histogram(
        x=r, histnorm='probability density', nbinsx=60,
        marker=dict(color=TEAL, line=dict(width=0.5, color=NAVY)),
        name=comm_names.get(col, col), showlegend=False,
    ), row=row, col=c)

    fig.add_trace(go.Scatter(
        x=x, y=pdf, mode='lines',
        line=dict(color=RED, width=1.8),
        name='Normal fit', showlegend=(idx == 0),
    ), row=row, col=c)

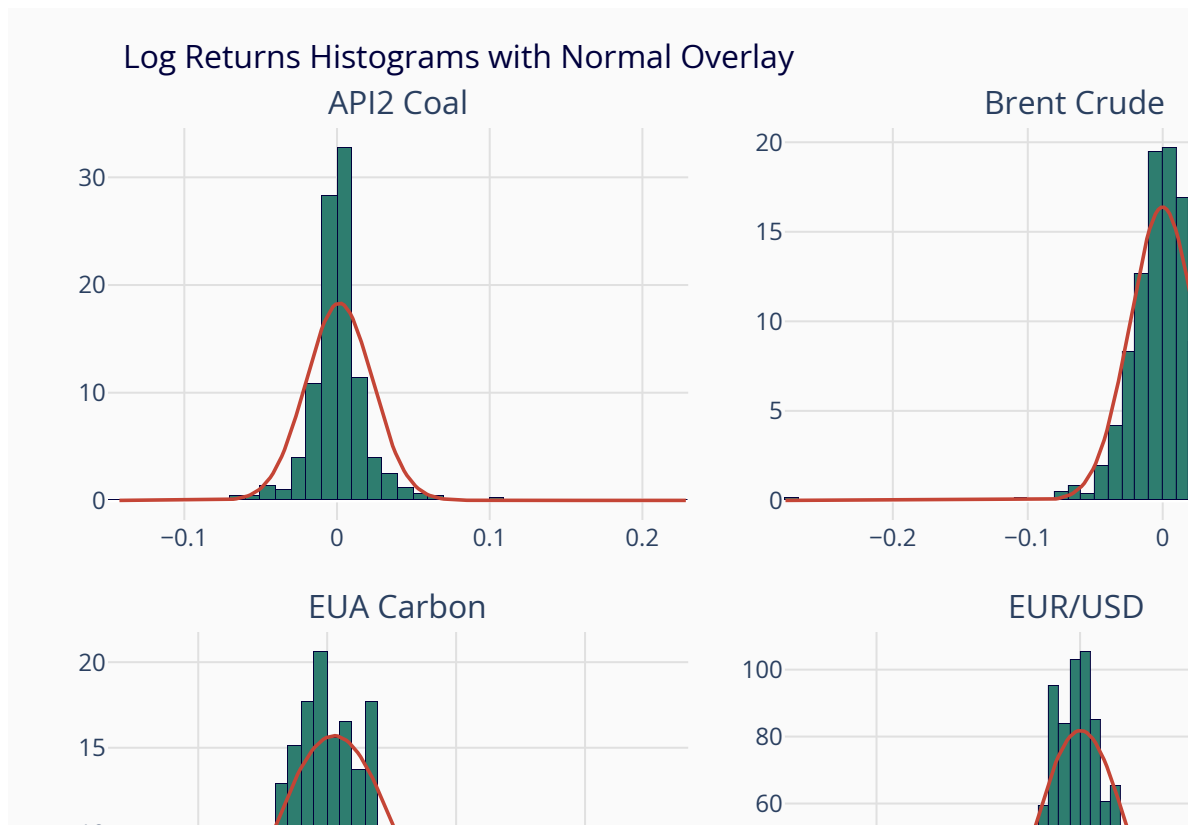
fig.update_layout(
    title=dict(text='Log Returns Histograms with Normal Overlay', font=dict(
        plot_bgcolor=OFFWHITE, paper_bgcolor=OFFWHITE,
        height=800, margin=dict(l=50, r=50, t=60, b=40),
```



```

)
fig.update_xaxes(gridcolor='#E0E0E0')
fig.update_yaxes(gridcolor='#E0E0E0')
fig.show()

```



TTF shows pronounced leptokurtosis, the histogram spikes higher at the centre and has heavier shoulders than the normal distribution allows. This is the statistical

signature of a market where daily moves of $\pm 5\%$ are routine and moves of $\pm 10\%$ occur several times per year. RBOB and coal show similar heavy-tailed behaviour, consistent with markets subject to supply disruptions and inventory cycles.

EUR/USD is the only series where the normal distribution provides a visually reasonable fit, and even here the Jarque-Bera statistic is elevated by the large sample size. For every other commodity the gap between the histogram bars and the red normal curve is the return mass a Gaussian VaR model misses. Notebook 4 recovers it by drawing from empirical GARCH residual quantiles instead of from a normal distribution.

6. Rolling volatility

Annualised volatility computed as $\sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{252}$ over a trailing 60-business-day window. The 60-day window is a desk convention: long enough to smooth out daily noise, short enough to reflect the current volatility regime.

A 20% annualised threshold is overlaid as a reference line. Sustained readings above this level typically trigger position reductions or additional hedging at institutional trading desks.

```
In [5]: rolling_vol = log_returns.rolling(60).std() * np.sqrt(252)

fig = go.Figure()
for i, col in enumerate(rolling_vol.columns):
    fig.add_trace(go.Scatter(
        x=rolling_vol.index, y=rolling_vol[col],
        mode='lines', name=comm_names.get(col, col),
        line=dict(color=COLORS[i % len(COLORS)], width=0.9),
    ))

fig.add_hline(y=20, line_dash='dash', line_color='#999999',
              annotation_text='20% threshold', annotation_position='right')

fig.update_layout(
    title=dict(text='Rolling 60-Day Annualised Volatility', font=dict(color=
    xaxis=dict(title='', gridcolor='#E0E0E0'),
    yaxis=dict(title='Annualised Volatility', gridcolor='#E0E0E0', ticksuffi
    plot_bgcolor=OFFWHITE, paper_bgcolor=OFFWHITE,
    legend=dict(orientation='h', y=-0.25),
    height=550, margin=dict(l=50, r=50, t=50, b=100),
    hovermode='x unified',
)
fig.show()
```



TTF's rolling volatility spends most of the sample above every other commodity, often exceeding 40% annualised. Three regimes are visible: a pre-2022 period with vol in the 20–35% band, a sharp spike around mid-2022 (the gas crisis), and a post-2023 normalisation to 15–25% as LNG imports and storage refills stabilised the market. Coal (API2) is the second-most volatile commodity, followed by RBOB gasoline, both are markets where inventory dynamics and supply disruptions produce episodic volatility clusters.

Carbon shows a steady volatility increase from 2020 onward, consistent with the tightening EU ETS cap under Phase IV. Brent and EUR/USD are the calmest series throughout. The 2022 spike in TTF volatility corresponds to the period when TTF traded from roughly €70 to €340/MWh and back within six months. Scaled by the pre-2022 volatility regime of 20–35% annualised, a cumulative move of that size over six months sits around seven standard deviations, which is the practical argument for a conditional volatility model: an unconditional estimate fitted across the whole sample would have called this impossible on the day it began.

7. Stationarity tests

The Augmented Dickey-Fuller (ADF) test evaluates the null hypothesis that a unit root is present, i.e., that the price series is non-stationary. For energy commodities, price levels are typically non-stationary (they do not revert to a fixed mean), but log returns should be stationary. This matters because most econometric models, including the GARCH and DCC specifications in Notebooks 3 and 4, assume stationary input series.

```
In [6]: from statsmodels.tsa.stattools import adfuller

print("Augmented Dickey-Fuller test (H0: unit root / non-stationary)")
print(f"{'Commodity':<20s} {'Level ADF':>10s} {'Level p':>10s} {'Return ADF':>10s} {'Return p':>10s}")
print("-" * 65)
for col in wide.columns:
    lev = adfuller(wide[col].dropna(), maxlag=20, autolag='AIC')
    ret = adfuller(log_returns[col].dropna(), maxlag=20, autolag='AIC')
    lev_v = "NR" if lev[1] > 0.05 else "S"
    ret_v = "S" if ret[1] < 0.05 else "NR"
    print(f"{'comm_names.get(col, col)':<20s} {lev[0]:>10.2f} {lev[1]:>10.4f} {ret[0]:>10.2f} {ret[1]:>10.4f}")
```

```
Augmented Dickey-Fuller test (H0: unit root / non-stationary)
Commodity          Level ADF    Level p Return ADF    Return p
-----
API2 Coal          -1.57      0.5009    -15.61    0.0000 (Level:NR
Return:S)
Brent Crude        -1.95      0.3072     -6.04    0.0000 (Level:NR
Return:S)
DE Baseload        -3.40      0.0111    -13.13    0.0000 (Level:S R
eturn:S)
EUA Carbon         -1.99      0.2932    -29.83    0.0000 (Level:NR
Return:S)
EUR/USD            -1.80      0.3801    -28.57    0.0000 (Level:NR
Return:S)
ULSD Heating Oil   -1.54      0.5161     -6.18    0.0000 (Level:NR
Return:S)
Nordic N01         -3.62      0.0053    -20.17    0.0000 (Level:S R
eturn:S)
RBOB Gasoline      -1.90      0.3300     -6.59    0.0000 (Level:NR
Return:S)
TTF Gas            -2.24      0.1915    -16.32    0.0000 (Level:NR
Return:S)
```

For every commodity in the panel, price levels fail to reject the unit root null; this is expected for traded assets. Log returns uniformly reject the unit root null at the 1% level, confirming stationarity. The GARCH and copula models in Notebooks 3 and 4 are therefore applied to these stationary return series, satisfying the distributional assumptions of the estimators.

8. Findings

1. **TTF gas is the most volatile commodity in the complex**, with a coefficient of variation above 100% and rolling vol routinely exceeding 30%. Gas markets lack global fungibility; regional supply shocks transmit directly into price.
2. **Every commodity except EUR/USD rejects normality** in a Jarque-Bera test. The excess kurtosis visible in the return distributions means a Gaussian VaR model systematically understates tail risk. Notebook 4 handles this in the marginals, by simulating from empirical GARCH residual quantiles.
3. **Price paths diverge over the sample**. German power and carbon end above where they started, while coal and gas trade below starting levels for most of the period. The gaps between these paths are the spreads that Notebook 2 decomposes into fuel and carbon cost.
4. **Returns are stationary**. The ADF test confirms that log-return series are suitable for the GARCH and DCC models that follow.
5. **Volatility clusters are regime-dependent**. TTF volatility tripled during the 2022 gas crisis relative to pre-2022 levels. A constant-volatility model calibrated before the crisis would have understated risk throughout it, and one calibrated across the full sample overstates risk in the calm years either side.

The next notebook examines how these commodities interact through the three core cross-commodity spreads: spark, dark, and crack.

References

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- Said, S.E. & Dickey, D.A. (1984). "Testing for unit roots in autoregressive-moving average models of unknown order." *Biometrika*, 71(3), 599–607.

PDF export

To export this notebook as a PDF for offline reading, run the cell below. Requires a LaTeX installation (`texlive-xetex` recommended) and `nbconvert` :

```
pip install nbconvert pandoc
sudo apt install texlive-xetex texlive-latex-extra
```

```
In [7]: # Uncomment to export PDF:
# !jupyter nbconvert --to pdf --template classic --output-dir ../docs/notebooks
print("PDF export: uncomment the line above and run to generate docs/notebooks/01_market_landscape.pdf")
```

PDF export: uncomment the line above and run to generate docs/notebooks/01_market_landscape.pdf